agentgate v1.0.0 Release Notes

Release date: 30 October 2026 | Licence: MIT

Summary

agentgate is a command-line tool for testing AI agents before production. It follows an 8-step framework (define criteria, build a golden dataset, test in layers, judge quality, decide at a release gate, release gradually, feed findings back). You point one config file at your agent's API, run one command, and get a one-page PDF report with a clear verdict.

This is the first public release.

Highlights

- **One-page report.** Every run writes `report.pdf` (always one page) and a full `results.json`.
- **Clear verdicts.** The release gate returns GO, NO-GO or PENDING MANUAL CHECKS. A smoke run returns SMOKE PASS or SMOKE FAIL.
- **Works with your agent.** An API config file (`config/agent.yaml`) maps your agent's request and response. A Python function can be used instead.
- **Two test tiers.** Smoke runs deterministic checks on every pull request. Full runs all cases, repeated runs and the LLM judge before release.
- **Zero-tolerance rules.** Must-never events (forbidden tools, leaked secrets) and security cases block a release regardless of percentages.
- **Judge with any model.** The LLM judge uses any OpenAI-compatible endpoint, so a local server or a hosted API (including Qwen) can be used.

What is included

Area	Contents
Criteria (step 1)	`config/thresholds.yaml` : frozen per-category thresholds, must-never rules, limits

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Dataset (step 2)	`dataset/golden_cases.json` (14 example cases, fake data only) and `agentgate validate`
Deterministic tests (steps 3-4)	Checks on tool choice, arguments, order, repeated calls, final state, handoff, token and call budgets
Judge (step 5)	Rubric-based judge, runs only after deterministic checks pass, cached results, `agentgate calibrate`
Security and limits (step 6)	Prompt-injection and leak cases, p95 latency and error-rate limits
Release gate (step 7)	Per-category thresholds, zero must-never events, manual rollback confirmation
Feedback loop (step 8)	`agentgate triage` anonymises and labels production findings before review
Automation	GitHub Actions workflow that runs validation, tests and the smoke tier
Demo	A rule-based demo agent, plus a deliberately flawed mode to see a NO-GO report

Commands

- `agentgate validate` checks the dataset.
- `agentgate run --tier smoke` runs the fast tier.
- `agentgate run --tier full` runs everything, with `--no-judge` and `--rollback-rehearsed` options.
- `agentgate calibrate` checks the judge against labelled examples.
- `agentgate triage` queues a production finding for review.
- `agentgate report` rebuilds a PDF from a `results.json`.

The exit code is 1 on NO-GO or SMOKE FAIL, so CI can block a merge.

Requirements

- Python 3.10 or newer
- Packages installed automatically: requests, pyyaml, reportlab
- For the judge: a reachable OpenAI-compatible endpoint and model name

Quick start

1. Unzip the project to a stable folder and open PowerShell (or a terminal) inside it.
2. Run `python -m venv .venv`, activate it, then `pip install -e ".[dev]"`.
3. Run `agentgate run --tier smoke --agent config/agent.demo.yaml`.
4. Open `reports/latest/report.pdf`.

Verification

- The project's own test suite (20 tests) passed in the build environment (Linux).
- The HTTP agent adapter, judge, calibration and triage were exercised against stand-in local servers.
- Both a passing run (GO) and a failing run (NO-GO) were checked, and each report fits on one page.

Known limitations

- Not tested against a real Qwen model or any real production agent. The judge was tested with a stand-in OpenAI-compatible server.
- Not tested on Windows by the author; the setup steps use standard Python tooling.
- Qwen3.7-Plus is listed by public model trackers as a hosted, API-only model. Using it as the judge sends test inputs and agent replies to that provider, so keep real customer data out of the dataset.
- agentgate does not replace a human red team, real load testing, a rehearsed rollback or domain-expert review.
- The example dataset is small, so percentages are noisy and the report shows a sample-size warning. Replace the examples with your own cases.
- Rollback rehearsal is a manual confirmation flag; the tool cannot verify it.

Notes for users

- Keep secrets in `.env` or CI secrets. `.env` is excluded by `.gitignore`.
- Changing `config/thresholds.yaml` changes the thresholds hash printed in every report, so edits are visible in review.
- Every report records the dataset hash, thresholds hash, judge model and prompt version, and commit, so a result can be traced.